

Gradient Starvation in Recurrent Networks: A Causal Cross-Entropy Field Theory

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Abstract

Gradient starvation (GS) occurs when a predictive feature suppresses the learning of another useful feature under cross-entropy training. Existing theory establishes GS in linearized settings, while recent work develops rich-regime dynamical field theories and reduced learning dynamics for recurrent networks. The missing intersection is a causal, optimization-time theory of *strong-weak feature competition in recurrent learning*. We propose a deliberately narrow ICLR-scale program: (i) derive exact cross-entropy projected dynamics for two task modes; (ii) obtain an exact finite-width reduction for a linear RNN; (iii) close the corresponding strong-weak dynamics with a joint recurrent DMFT and a matched weak-only counterfactual; (iv) predict a starvation phase boundary as a function of feature-strength ratio and temporal separation; and (v) use the identified interaction to design a separate physics-inspired mitigation. This document fixes the paper’s scope, theorem package, experiments, and completion checklist. The central DMFT result is explicitly marked as a target until its concentration/closure proof is complete.

1 Introduction

Cross-entropy training can stop learning a useful feature even when that feature is predictive. Pezeshki et al. [1] formalized this phenomenon as *gradient starvation*: learning a stronger feature can reduce the learned response to a weaker one. Their analysis, however, is tied to a linearized/NTK-style setting. In recurrent networks the problem is richer: a feature can be weak because of its statistical strength, because it occurs at a difficult temporal lag, or because a dominant feature actively suppresses its gradient. A convincing recurrent theory must distinguish these causes.

Our central principle is therefore **causal**: the weak feature is starved only if it learns more slowly when the strong feature is present than in a matched weak-only task with the same weak signal, lag, noise, architecture, and optimization protocol. We track feature modes rather than individual neurons or scalar weights. This matters because recent RNN learning theories already show that low-dimensional task structure can be distributed across many units and that learning dynamics can require additional “loss-invisible” state variables [3].

The paper is intentionally narrow enough to fit a 9-page ICLR submission. The main text will make three claims: **(C1)** exact cross-entropy strong-to-weak suppression laws; **(C2)** a recurrent DMFT that predicts a causal starvation phase boundary; and **(C3)** a separate theory-guided interaction regularizer that reduces the predicted starvation mechanism. CNN/ANN transfer is supporting evidence, not a second theory paper.

2 Current Landscape and the Novelty Gap

Gradient starvation itself is not new: Pezeshki et al. [1] prove that increasing one feature’s strength can suppress another under cross-entropy and introduce Spectral Decoupling as mitigation. Rich-regime feature learning is also not new: Bordelon and Pehlevan [2] derive self-consistent dynamical field theory (DFT) for evolving activation and gradient kernels in wide networks. In recurrence, Ger and Barak [3] derive exact reduced learning ODEs for linear low-rank RNNs and asymptotically exact nonlinear dynamics under a large- N Gaussian regime. Bauer et al. [4] develop a unified mean-field view of feature learning in

RNNs/DNNs, and Clark et al. [5] derive a task-conditioned RNN DMFT in which disorder coexists with learned structure.

Hence we should *not* claim “first DMFT for trained RNNs,” “first rich-regime RNN feature theory,” or that GS breaks microscopic exchangeability. The defensible novelty is the intersection:

A causal cross-entropy field theory of recurrent gradient starvation that jointly models strong and weak temporal modes, separates logistic margin-gating from learned recurrent geometry, solves a matched weak-only counterfactual, and predicts a starvation phase boundary and observable GSI statistics.

This also resolves the role separation in the project: **GSI measures**, **DMFT explains/proves**, and a **regularizer mitigates**. The proof and the intervention are separate contributions.

3 Scope, Setup, and Causal Definition

We first study a binary two-feature teacher. Let z_s, z_w denote fixed signed strong/weak task coordinates and

$$M(x; \tau) = y f_{\theta(\tau)}(x) = m_s(\tau) z_s(x) + m_w(\tau) z_w(x), \quad (1)$$

with logistic loss $\ell(M) = \log(1 + e^{-M})$. Define

$$w(M) = \sigma(-M), \quad q(M) = \sigma(M)\sigma(-M) > 0. \quad (2)$$

For sequence tasks the feature coordinates are projections onto fixed temporal filters ψ_s, ψ_w . Their strength ratio is ρ and their lag separation is $\Delta_\ell = |\ell_s - \ell_w|$. Sequence time t and optimization time τ are distinct throughout.

Definition 1 (Causal gradient starvation). Let $m_w^B(\tau)$ be the weak-mode response in the *both-feature* task and $m_w^W(\tau)$ the response in the matched *weak-only* task. For a target response β , define

$$\Delta T_w(\beta) = T_w^B(\beta) - T_w^W(\beta), \quad T_w(\beta) = \inf\{\tau : m_w(\tau) \geq \beta\}. \quad (3)$$

We call the weak mode causally starved when $\Delta T_w(\beta) > 0$ on a nontrivial target range. Instantaneous suppression is measured by the difference in weak drift at matched weak state.

Controlled recurrent model. The main theorem uses a linear RNN $h_{t+1} = W h_t + b x_t$, $f = c^\top h_T$, with a random bulk plus rank-2 task structure,

$$W(\tau) = \frac{g}{\sqrt{N}} J + \frac{1}{N} [u_s(\tau) v_s(\tau)^\top + u_w(\tau) v_w(\tau)^\top], \quad (4)$$

where $J_{ij} \sim \mathcal{N}(0, 1)$ is quenched. The first theorem version uses full-batch cross-entropy gradient flow and trains only the explicitly stated structured/input/readout variables. Nonlinear RNNs are a validation extension, not part of the first rigorous claim.

4 Exact Theorem Package

4.1 Architecture-independent cross-entropy flow

Theorem 1 (Exact signed function-space flow; PROVED). *For any differentiable network trained by full-batch gradient flow,*

$$\dot{\theta} = \mathbb{E}_{x'} [w(M(x')) y' \nabla_\theta f_\theta(x')], \quad (5)$$

$$\dot{M}(x) = \mathbb{E}_{x'} [\widetilde{K}_\theta(x, x') w(M(x'))], \quad (6)$$

where $\widetilde{K}_\theta(x, x') = y y' \langle \nabla_\theta f_\theta(x), \nabla_\theta f_\theta(x') \rangle$.

Proof. Since $\ell'(M) = -\sigma(-M) = -w(M)$ and $\nabla_\theta M = y \nabla_\theta f_\theta$, gradient flow gives the first identity. Multiplying by $\nabla_\theta M(x)^\top$ yields the second. No fixed-kernel approximation is used. $\square$

4.2 Exact two-mode rich-regime dynamics

Define

$$g_a(m_s, m_w) = \mathbb{E}[z_a w(M)], \quad G_{ab}(\theta) = \langle \nabla_\theta m_a, \nabla_\theta m_b \rangle. \quad (7)$$

Theorem 2 (Exact projected flow; PROVED). *If the two-mode margin representation is exact, then for finite parameter dimension*

$$\dot{m}_a = \sum_{b \in \{s, w\}} G_{ab}(\theta) g_b(m_s, m_w). \quad (8)$$

Proof. $\partial L / \partial m_b = -g_b$, so $\nabla_\theta L = -\sum_b g_b \nabla_\theta m_b$. Taking the inner product of $-\nabla_\theta L$ with $\nabla_\theta m_a$ gives the claim. Thus g captures loss/data competition and G captures evolving parameter geometry. $\square$

4.3 Strong-to-weak suppression and feature-strength starvation

Let $A_{ab} = \mathbb{E}[z_a z_b q(M)]$.

Theorem 3 (Cross-feature CE sensitivity; PROVED).

$$\frac{\partial g_w}{\partial m_s} = -A_{sw}, \quad g_w(m_s, m_w) - g_w(0, m_w) = -\int_0^{m_s} A_{sw}(a, m_w) da. \quad (9)$$

Hence $A_{sw} > 0$ on the relevant path is a sufficient condition for direct strong-to-weak suppression.

Proof. Differentiate $g_w = \mathbb{E}[z_w \sigma(-M)]$ and use $d\sigma(-M)/dM = -q(M)$; integration gives the second identity. $\square$

For an equilibrium statement, scale the strong coordinate as $M = \rho m_s \bar{z}_s + m_w z_w$ and consider

$$J_\rho = \mathbb{E}[\ell(M)] + \frac{\lambda}{2}(m_s^2 + m_w^2), \quad \lambda > 0. \quad (10)$$

Theorem 4 (Feature-strength starvation; PROVED). *At the unique optimum, if $m_s^* > 0$ and $A_{sw}^* > 0$, then*

$$\frac{dm_w^*}{d\rho} = -\frac{2\lambda m_s^* A_{sw}^*}{\det H} < 0, \quad (11)$$

where $H = \begin{pmatrix} \lambda + \rho^2 A_{ss} & \rho A_{sw} \\ \rho A_{sw} & \lambda + A_{ww} \end{pmatrix}$ is the positive-definite Hessian.

Proof. The stationarity conditions are $F_s = \lambda m_s - \rho \bar{g}_s = 0$ and $F_w = \lambda m_w - g_w = 0$. By the implicit-function theorem, $dm^*/d\rho = -H^{-1} \partial_\rho F$. The weak component simplifies to $-A_{sw}(\rho \bar{g}_s + \lambda m_s)/\det H$; use $\rho \bar{g}_s = \lambda m_s$ at the optimum. $\square$

4.4 Causal trajectory theorem

Theorem 5 (Weak-only comparison; PROVED under common-metric closure). *Assume $\dot{u} = \kappa_w(\tau) g_w(m_s(\tau), u)$ and $\dot{v} = \kappa_w(\tau) g_w(0, v)$, with $\kappa_w > 0$, $u(0) = v(0)$, and $A_{sw} > 0$ on the reachable region. Then $u(\tau) \leq v(\tau)$ for all τ , hence $\Delta T_w(\beta) \geq 0$ whenever both trajectories reach β .*

Proof. At any matched weak state, the sensitivity theorem gives $g_w(m_s, m) \leq g_w(0, m)$. At a hypothetical first upward crossing $u = v$, one has $\dot{u} \leq \dot{v}$, so the crossing is impossible. The hitting-time inequality follows. $\square$

5 Exact Linear-RNN Reduction and Main DMFT Target

For the linear RNN, define the impulse response $r_\ell = c^\top W^\ell b$ and temporal-mode response $m_a = \sum_\ell \psi_a(\ell) r_\ell$. Let

$$p_a = \sum_\ell \psi_a(\ell) W^\ell b, \quad q_a = \sum_\ell \psi_a(\ell) (W^\ell)^\top c, \quad (12)$$

$$S_a = \sum_\ell \psi_a(\ell) \sum_{k=0}^{\ell-1} (W^k)^\top c [W^{\ell-1-k} b]^\top. \quad (13)$$

Theorem 6 (Finite- N linear-RNN geometry; PROVED). *The exact projected dynamics are $\dot{m}_a = \sum_b G_{ab} g_b$ with*

$$G_{ab} = p_a^\top p_b + q_a^\top q_b + \langle S_a, S_b \rangle_F. \quad (14)$$

Proof. Solve $h_T = \sum_t W^{T-1-t} b x_t$ and differentiate $r_\ell = c^\top W^\ell b$. The identity $d(W^\ell) = \sum_{k=0}^{\ell-1} W^k (dW) W^{\ell-1-k}$ gives the three parameter-block gradients. Their Euclidean Gram inner product is exactly G_{ab} , and the projected-flow theorem completes the result. $\square$

Main theorem to complete before submission (Target). As $N \rightarrow \infty$ at fixed sequence horizon and finite learning horizon, the random-bulk rank-2 RNN should converge to a deterministic joint strong-weak DMFT with recurrent correlation/response order parameters and

$$\dot{m}_a = \sum_b G_{ab}(\tau) \mathbb{E}[z_b \sigma(-(m_s z_s + m_w z_w))]. \quad (\text{DMFT})$$

The same self-consistency system with the strong teacher coordinate removed defines the weak-only counterfactual. Crucially, the derivation must retain cross-order parameters such as G_{sw} (and the corresponding projected correlation/response objects); two independent DMFTs would delete the interaction to be explained.

A reviewer-safe proof must complete six obligations: **(i)** exact scaling/probability space; **(ii)** quenched-disorder average by generating functional or cavity method; **(iii)** effective single-site stochastic process; **(iv)** closure with all required visible/invisible order parameters; **(v)** finite-horizon existence/uniqueness and concentration/self-averaging; and **(vi)** matched weak-only convergence sufficient to pass to hitting-time delay. If only a formal saddle point is available for a nonlinear extension, it must be labeled a DMFT-limit prediction rather than a universal theorem.

Starvation susceptibility and phase boundary. For $F_w = \dot{m}_w = G_{ws} g_s + G_{ww} g_w$, the fixed-geometry contribution is

$$\left. \frac{\partial F_w}{\partial m_s} \right|_G = -G_{ws} A_{ss} - G_{ww} A_{sw}. \quad (15)$$

The full rich-regime susceptibility additionally contains $\partial G / \partial m_s$, supplied by DMFT response functions. Define

$$\chi_{w \leftarrow s} := -\frac{\partial F_w}{\partial m_s}. \quad (16)$$

The flagship theoretical result is a surface $\rho_c(\Delta_\ell, g)$ separating jointly learned, delayed-weak, and strongly starved regimes, verified against the causal ΔT_w criterion.

6 Experiments: Only Three Main Blocks

The main paper should run only experiments that test the three claims. All larger sweeps and additional datasets go to the appendix.

Table 1: Minimal main-text experiment budget. Larger sweeps belong in the appendix.

Block	Purpose	Main output
E1	Establish causal GS and map $\rho \times \Delta_\ell$	Phase diagram + weak-only control
E2	Validate the DMFT quantitatively	Trajectories + width convergence
E3	Show theory-guided mitigation and one transfer result	Interaction reduction + robustness

6.1 E1: Causal synthetic phase diagram

Generate binary sequence data with two fixed predictive temporal features. Use $T = 20$, a linear RNN of width $N = 512$, full-batch BCE, and 5 seeds. Sweep

$$\rho \in \{1, 2, 4, 8\}, \quad \Delta_\ell \in \{0, 2, 4, 8\}, \quad (17)$$

and run both-feature and matched weak-only conditions at every point. Log m_s, m_w , their drifts, margins, A_{sw} , G_{ab} , loss, and GSI-5. Primary metric: $\Delta T_w(\beta)$ at a preregistered β (plus area between weak trajectories as a robustness metric). **Main Figure 1:** heatmap of causal starvation in (ρ, Δ_ℓ) with the theoretical boundary overlaid. Include one negative-control data regime with $A_{sw} \leq 0$; starvation should not be automatic merely because ρ is large.

6.2 E2: DMFT-to-network agreement

At four representative phase-diagram points, solve the joint DMFT and compare to finite networks at $N \in \{128, 512, 2048\}$ (3–5 seeds). Compare $m_s(\tau)$, $m_w(\tau)$, margin distribution, G_{ss}, G_{sw}, G_{ww} , and

$$\text{GSI}_5 = 1 - \frac{N_{\text{eff}}}{N_{\text{ex}}}, \quad N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \quad (18)$$

Report trajectory RMSE and width-scaling of DMFT error. First verify numerically at finite N that the exact identity $\dot{m} = Gg$ matches autograd to integration precision. **Main Figure 2:** theory/empirical trajectories plus width-convergence inset. This experiment is what earns the use of DMFT over a purely two-dimensional ODE.

6.3 E3: Mitigation and limited transfer

Only after E1–E2 validate the mechanism, introduce a separate physics-inspired interaction penalty. The preferred target is the *negative starvation susceptibility*; a differentiable prototype is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_R [\max(0, \hat{\chi}_{w \leftarrow s})]^2, \quad (19)$$

with a first-order proxy based on the empirically estimable A_{sw}/G_{sw} terms if second-order differentiation is too expensive. Compare **ERM vs Spectral Decoupling vs Ours**. Validate on: (i) the synthetic linear RNN, (ii) one tanh-RNN or GRU variant, and (iii) one real spurious-feature benchmark (Waterbirds is the default). Primary outcomes: reduced ΔT_w , reduced GSI-5, preserved average accuracy, and improved worst-group accuracy on the real benchmark. **Main Figure 3:** mechanism reduction and downstream robustness. Do not treat the regularizer as part of the proof.

7 Proposed Contributions and Main-Paper Story

The final paper should claim exactly the following, conditional on completion of the target theorem and experiments:

1. **Causal definition and exact CE mechanism.** We separate true starvation from ordinary temporal-credit difficulty using matched weak-only training and derive exact strong-to-weak CE suppression identities.

2. **Recurrent theory.** We derive exact finite-width two-mode linear-RNN dynamics and a joint optimization-time recurrent DMFT that predicts the strong/weak geometry and starvation phase boundary.
3. **Observable bridge.** The theory predicts weak-mode learning curves, margin statistics, and GSI-5, tying the mathematical mechanism to measurements on trained networks.
4. **Theory-guided mitigation.** A separate interaction regularizer suppresses the mechanism identified by the theory and improves weak-feature/robust performance without simply destroying the dominant feature.

The narrative is therefore

Define $\rightarrow$ Prove $\rightarrow$ Predict a phase boundary $\rightarrow$ Validate $\rightarrow$ Mitigate.

This is preferable to a broader “RNN + ANN + CNN + five diagnostics” paper, which would dilute the central theorem.

8 Final ICLR-27 Oral-Ready Checklist

Theory	☐ Freeze exact model/scaling and probability space. ☐ Keep sequence and optimization time distinct. ☐ Complete the joint CE-DMFT derivation and weak-only corollary. ☐ Prove closure or add missing loss-invisible variables. ☐ Establish finite-horizon concentration or state the precise non-rigorous limit. ☐ Derive $\chi_{w \leftarrow s}$ and $\rho_c(\Delta_\ell, g)$.
Validation	☐ E1 gives a reproducible causal phase diagram. ☐ Negative control obeys the theorem’s sign condition. ☐ Exact finite- N $\dot{m} = Gg$ identity matches autograd. ☐ E2 shows DMFT error decreases with width. ☐ Predicted margin law/GSI-5 match measurements.
Mitigation	☐ Regularizer targets the derived interaction, not an arbitrary weight statistic. ☐ Compare ERM/SD/Ours. ☐ Reduce ΔT_w and GSI-5 while preserving dominant-feature performance. ☐ Include one nonlinear recurrent and one real-data transfer result.
Statistics	☐ 5 seeds for central synthetic plots; ≥ 3 elsewhere. ☐ 95% CIs/effect sizes. ☐ Predeclare the primary β , phase criterion, and hyperparameter-selection rule. ☐ Report compute and sensitivity to λ_R .
Paper	☐ One central theorem, one phase diagram, one DMFT-vs-network figure, one mitigation figure. ☐ Put full generating-functional algebra and secondary sweeps in appendix. ☐ Explicitly distinguish proven, conditional, and empirical claims. ☐ Include limitations/falsification regimes. ☐ Release code, seeds, configs, and synthetic generator. ☐ Run an adversarial reviewer pass against every novelty and theorem statement.

Suggested 9-page submission allocation. Introduction + landscape: 1.5 pages; setup/contributions: 0.75; theorem statements and proof sketches: 2.75; DMFT result/phase law: 1.25; experiments: 2.0; discussion/limitations: 0.5; references excluded/handled by venue style. Full proofs, derivation algebra, additional GSI variants, extra widths, ANN/CNN datasets, and hyperparameter sweeps go to the appendix.

9 Conclusion

The paper should not try to prove every form of gradient starvation in every architecture. Its strongest oral-level version is a precise recurrent statement: a dominant temporal feature changes the cross-entropy learning field available to a weaker predictive feature; this effect is causally identified by a weak-only counterfactual, quantitatively closed by a joint recurrent DMFT, and summarized by a phase boundary. A separate interaction regularizer then tests whether the theory identifies a mechanism that can actually be controlled. The current exact theorems provide the finite-dimensional backbone; the decisive remaining mathematical task is the joint CE-DMFT concentration/closure theorem, and the decisive empirical task is agreement between that theory and the causal (ρ, Δ_ℓ) phase diagram.

References

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